Relational Geometry and the Emergence of Gravity

From Harmonic Closure to Covariant Relativity

Draf v5 —14 April 2026

Luis Diego Mata Sánchez
Cartago, Costa Rica

Manuscript status. This is the complete synthesis of the relational framework to date. All claims carry explicit labels (Table 1). Version 9.1 incorporates three critical revisions over v9.0: (i) the derivation of $\eta(n) = \alpha I(n)$ is now carried out entirely in terms of modal invariants and the coupling α, without direct reference to G; (ii) claims about fermionic realizations and black-hole information are correctly marked **[SPECULATIVE]**; (iii) the algebraic–physical dictionary (Table 5) is complete. The framework remains exploratory and falsifiable; no experimental validation independent of the author's prior program is claimed.

Table 1: Claim-type taxonomy.

Label	Meaning
[POSTULATE]	Foundational assumption — not derived, not yet falsified.
[DERIVED]	Mathematical consequence of stated postulates.
[CALIBRATION]	Parameter fixed by matching one empirical value.
[HYPOTHESIS]	Testable physical conjecture requiring independent validation.
[SPECULATIVE]	Exploratory program note; no prediction claimed.
[OPEN]	Precisely stated unresolved problem.

Contents

1 Introduction **4**

2 Mathematical Core: Generative Algebra **5**
2.1 Primitive elements . . . 5
2.2 Generated hierarchy . . . 5
2.3 Quaternionic realisation . . . 5
2.4 The modal invariant . . . 6
2.5 Structural motivation for $n = 4$ . . . 7
2.6 The arithmetic witness: $2^n = 1 + (2^n - 1)$ . . . 7

3 Algebra–Physics Dictionary **8**
3.1 Calibration . . . 8
3.2 Gravitational instability parameter . . . 8

4 Non-relativistic Physical Framework **10**
4.1 Relational distance and the inverse-square law . . . 10
4.2 Gravitational potential . . . 10
4.3 Calibration . . . 10
4.4 Natural gravitational scale . . . 10
4.5 Three centripetal accelerations of $n = 4$ . . . 11
4.6 Geometric relation . . . 11

5 Covariant Extension **12**
5.1 Scalar field and covariant realisation . . . 12
5.2 Action and field equations . . . 12
5.3 Newtonian limit . . . 12
5.4 Cosmological consequences . . . 12

6 The Open Problem of α_G **13**
6.1 Reduction to a single parameter . . . 13
6.2 Three routes — all fail . . . 13
6.3 The E_8 program and its status . . . 13
6.4 A numerical hint . . . 13

7 Testable Predictions **15**
7.1 Weak equivalence principle violations . . . 15
7.2 Cosmological deviations from ΛCDM . . . 15
7.3 Scale-dependent gravity . . . 15

8 Discussion **16**
8.1 Comparison with emergent-gravity frameworks . . . 16
8.2 The mechanism . . . 16
8.3 Other manifestations . . . 16
8.4 Limitations and future work . . . 16

9 Conclusion **17**

A Contraction of $P_{\mu\nu}$ **18**

B Geometric relation $R_V^2 = I(n)R_N R_m$ **18**

C Generator matrices and commutator verification **18**

D Glossary of symbols **18**

Abstract

We present a unified relational framework in which gravity emerges as the macroscopic tendency of physical systems to reduce the phase offset between their projected relational states. The core is a generative algebra over a real normed vector space with an antisymmetric operator $\times$ satisfying restricted norm compatibility, metric compatibility, and double-projection closure with binary orientation $\sigma \in \{+1, -1\}$. This algebra is explicitly realised as the quaternionic cross product on $\mathrm{Im}(\mathbb{H})$; one orientation yields $\mathfrak{so}(3)$, both orientations yield $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$, identifying σ as the algebraic distinction between left and right chiral sectors.

The four generated modes $\{U, V, m, a\}$ acquire norms R, R^2, $R\sqrt{1-R^2}$, R^3. Upon calibration with $c, \hbar$ and binary closure depth n, they satisfy $\omega_N^2 = \omega_V^2 + \omega_m^2$ with $\omega_V^2/\omega_N^2 = 2^{-n}$. The modal invariant $I(n) = 2^{n/2}\sqrt{2^n - 1}$ unifies all cross-relations: $G/\alpha = I(n)\,\hbar c/m_p^2$, $R_V^2 = I(n) R_N R_m$, and $(a_V/a_N)(a_V/a_m) = 1/I(n)$. For $n = 4$, $I(4)^2 = 240$, the number of roots of E_8.

The gravitational instability parameter is expressed purely in terms of modal invariants and the coupling α, without direct reference to G:

$$\eta(n) = \alpha\, I(n),$$

leading to $\alpha_G = \alpha I(4) = I(4)/I(n_c)$, where n_c is the Planck closure depth defined by $\eta(n_c) = 1$. The coupling α is subsequently fixed by one empirical calibration. A heuristic approximation $\alpha_G \approx 2^{-127}(1 + 1/240)$ follows from $n_c \approx 131 = 127 + 4$.

Falsifiable predictions: (i) WEP violations correlated with nuclear binding energy per nucleon; (ii) cosmological deviations from ΛCDM of order 1–3% at $z \sim 1$–2, testable by LSST, Euclid, and DESI. **Central open problems:** (1) determine n_c algebraically; (2) complete the E_8 chain (Steps C–E); (3) derive isotropy from the quaternionic structure.

1 Introduction

Gravity remains the only fundamental interaction whose coupling constant G is imposed empirically rather than derived from deeper principles. While General Relativity provides an exquisite classical description, its conceptual origin — why mass curves spacetime with precisely this strength — remains unexplained. The question this paper addresses is:

> *Can the value of G, and the inverse-square form of the gravitational force, be expressed as consequences of pre-existing microscopic structure, rather than treated as primitive inputs?*

The framework applies a harmonic partition principle previously used in three domains (Table 2). Here we extend it to gravity, adding: the inverse-square law, Newton's constant, a natural gravitational scale, an explicit algebraic classification of the generative operator, and a purely modal expression for the gravitational instability parameter.

Table 2: Prior applications of the relational framework. All are internal consistency results, not independent validations.

Scale	System	Harmonic ratio	Result
Hadronic	Proton charge radius	$\omega_V^2 : \omega_m^2 = 1 : 15$	$r_p \approx 0.8412$ fm
Atomic	Valence stability (s–p)	$a : b$ prime or 2^n	Weight $P = a + b$
Cosmological	Hubble expansion	$V = \omega R$	$H_0 = \omega_V$ at $R = R_H$
Gravitational	Newtonian	$n = 4$ (nucleon)	$G = \alpha\kappa\omega_m^2$, $1/r^2$ derived

Acknowledged limitations

Part I (non-relativistic): weak-field, stationary regime only; α is calibrated, not derived (one free parameter); isotropy of Postulate 5 is stated but not derived from the algebra; n-selection is motivated, not proved; prior applications are internal consistency results.

Part II (covariant extension): the matter coupling is postulated, not derived from the algebra; strong-field exact solutions not obtained; β-regularisation is speculative; α_G remains open pending determination of n_c.

2 Mathematical Core: Generative Algebra

2.1 Primitive elements

Postulate 1 (Reference state)**.** *Let $\mathcal{V}$ be a real normed vector space. There exist a normalised reference vector $\hat{\aleph} \in \mathcal{V}$ with $\|\hat{\aleph}\| = 1$, and a differentiation marker $R \in \mathcal{V}$ with $\|R\| = R \in [0, 1]$ and $\langle \hat{\aleph}, R\rangle = 0$. Define $\mathcal{V}_R := \{A \in \mathcal{V} \mid \langle A, R\rangle = 0\}$.*

Postulate 2 (Generative operator)**.** *There exists an antisymmetric bilinear operator $\times : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ satisfying, for all $A \in \mathcal{V}_R$:*

$$\|A \times R\| = \|A\|\,\|R\|, \qquad \text{(P1 – restricted norm compatibility)}$$

$$\langle A \times R,\, A\rangle = 0, \qquad \text{(P2 – metric compatibility)}$$

$$(A \times R) \times R = \sigma R^2 A, \quad \sigma \in \{+1, -1\}. \qquad \text{(P3 – double-projection closure)}$$

Remark 1. Hurwitz's theorem (1898) states that a normed composition algebra over $\mathbb{R}$ exists only in dimensions $1, 2, 4, 8$. The restricted form (P1) is weaker: it is exactly the defining property of the cross product on $\mathbb{R}^3$ and the adjoint action of $\mathfrak{su}(2)$.

Postulate 3 (Orthogonal decomposition)**.** *The first projection $U := \hat{\aleph} \times R$ admits a unique decomposition $U = V + m$ with $\langle m, V\rangle = 0$.*

2.2 Generated hierarchy

Definition 1 (Four modes)**.**

$$U := \hat{\aleph} \times R, \quad V := U \times R, \quad a := V \times R, \quad m := U - V.$$

Proposition 1 (Modal norms)**.** *From Postulates 1–3:*

$$\|U\| = R, \quad \|V\| = R^2, \quad \|m\| = R\sqrt{1 - R^2}, \quad \|a\| = R^3.$$

No fifth independent mode is generated.

Proof. $\|U\| = \|\hat{\aleph} \times R\| = \|\hat{\aleph}\|\|R\| = R$ by (P1). $V = (\hat{\aleph} \times R) \times R = \sigma R^2 \hat{\aleph}$ by (P3), so $\|V\| = R^2$. By Postulate 3 and Pythagoras: $\|m\|^2 = \|U\|^2 - \|V\|^2 = R^2(1 - R^2)$. Since $V \in \mathcal{V}_R$, $\|a\| = \|V\|\|R\| = R^3$ by (P1). Applying $\times R$ to a: $a \times R = (V \times R) \times R = \sigma R^2 V$, proportional to V — no new direction is generated. □ □

Table 3: Hierarchy of algebraic modes.

Mode	Norm	Role	Frequency at $n = 4$
U (arena)	R	Space of distinguishability	ω_N
V (visible)	R^2	Relational/transport channel	$\omega_N/4$
m (confinement)	$R\sqrt{1 - R^2}$	Identity/inertial channel	$\sqrt{15}\,\omega_N/4$
a (interaction)	R^3	Adjustment/force channel	ω_N

2.3 Quaternionic realisation

Theorem 1 (Single orientation: $\mathfrak{so}(3)$)**.** *The modes $\{V, m, a\}$ with operator $\times$, restricted to* $\mathrm{span}\{V, m, a\}$, *realise the imaginary quaternionic cross product on* $\mathrm{Im}(\mathbb{H})$ *under $V \leftrightarrow \mathbf{i}$, $m \leftrightarrow \mathbf{j}$, $a \leftrightarrow \mathbf{k}$. The left-multiplication generators satisfy*

$$[L_V, L_m] = 2L_a, \quad [L_m, L_a] = 2L_V, \quad [L_a, L_V] = 2L_m,$$

so $\mathrm{span}\{L_V, L_m, L_a\} \cong \mathfrak{so}(3) \cong \mathfrak{su}(2) \cong A_1$.

Proof. Explicit 4×4 real matrices on the basis $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$:

$$L_V = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad L_m = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad L_a = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

All three commutator identities are verified by direct matrix multiplication. The arena mode U acts trivially ($[L_U, L_i] = 0$), consistent with the real unit $1 \in \mathbb{H}$. □ □

Theorem 2 (Both orientations: $\mathfrak{so}(4)$)**.** *The right-multiplication generators $\{R_V, R_m, R_a\}$ (corresponding to $\sigma = +1$) satisfy*

$$[L_i, L_j] = 2L_k, \quad [R_i, R_j] = 2R_k, \quad [L_i, R_j] = 0 \quad \forall\, i, j \in \{V, m, a\}.$$

Hence $\mathrm{span}\{L_V, L_m, L_a, R_V, R_m, R_a\} \cong \mathfrak{so}(3)_L \oplus \mathfrak{so}(3)_R \cong \mathfrak{so}(4) \cong D_2$.

Proof. Right-multiplication matrices:

$$R_V = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad R_m = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad R_a = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

All six cross-commutators $[L_i, R_j]$ vanish by direct computation. Hence $\sigma = \pm 1$ is the algebraic distinction between left and right quaternionic action. □ □

Corollary 1 (Chiral structure)**.** *The decomposition $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ is the chiral decomposition of the Lorentz algebra in four dimensions. The two orientations $\sigma = \pm 1$ therefore correspond algebraically to left and right chiral sectors.* ***[SPECULATIVE]*** *A full realisation of Dirac fermions would require additional structure (Clifford algebra, Hilbert space, dynamics) and remains an open program.*

2.4 The modal invariant

Definition 2 (Binary closure depth)**.** A system achieves *binary closure at depth n* when $\omega_V^2/\omega_N^2 = 2^{-n}$, equivalently $\omega_V^2 : \omega_m^2 = 1 : (2^n - 1)$.

Definition 3 (Modal invariant)**.** $I(n) := (\omega_N/\omega_V)(\omega_m/\omega_V) = 2^{n/2}\sqrt{2^n - 1}$.

[DERIVED] Since $\gcd(2^n, 2^n - 1) = 1$:

$$I(n)^2 = 2^n(2^n - 1) = \mathrm{lcm}(2^n, 2^n - 1).$$

For $n = 4$:

$$\boxed{I(4)^2 = 16 \times 15 = 240 = \#\mathrm{roots}(E_8).}$$

This equality is arithmetically exact; its algebraic interpretation — whether the modal hierarchy embeds naturally into the E_8 root lattice — is an open program detailed in Section 6.3.

Table 4: Values of $I(n)$ and structural context.

n	$I(n)$	$I(n)^2$	$2^n - 1$	Physical context
1	1.414	2	1 (unit)	Reference
2	3.464	12	3 (prime)	Electron family
3	7.483	56	7 (prime)	Neutron (unstable)
4	**15.49**	**240**	3×5	Proton
5	31.50	992	31 (prime)	—
7	127.5	16256	127 (Mersenne prime)	Planck threshold?

(For $n = 7$, $I(7) = 8\sqrt{127} \approx 127.5$; $I(7)^2 = 128 \times 127 = 16256$.)

2.5 Structural motivation for $n = 4$

Two independent arguments converge on $n = 4$; both are motivations, not proofs.

(i) Arithmetic irreducibility. $2^4 - 1 = 15 = 3 \times 5$ is the first confinement ratio that factors into two distinct primes, making the two modes arithmetically independent.

(ii) Real dimension of $\mathfrak{su}(2)$. The fundamental representation of $\mathfrak{su}(2)$ is 4-dimensional over $\mathbb{R}$. Under the reading that n encodes the minimal real degrees needed for a coherent spin description, $n = 4$ is the minimal compatible depth. Theorem 1 corroborates this: the generated algebra is exactly $\mathfrak{su}(2)$.

2.6 The arithmetic witness: $2^n = 1 + (2^n - 1)$

[DERIVED] The closure condition rewrites as $2^n = 1 + (2^n - 1)$. In the quaternionic realisation: the $\mathbf{1}$ is the real unit $1 \in \mathbb{H}$ (arena mode U), the background that enables observation and never disappears from the algebra; the $2^n - 1$ is proportional to $\|m\|^2$ (confinement mode), the structure registered as the physical system. This explains why the proton's confinement ratio is 15 (not 16), and why atomic valence gives 3 (not 4).

[SPECULATIVE] As $n \to \infty$ the visible mode vanishes while the real unit 1 persists algebraically. This persistence offers a possible algebraic interpretation of black-hole information: information becomes inaccessible because the visible channel closes, but the algebraic identity element remains. A full resolution requires a quantum treatment and is not claimed here.

3 Algebra–Physics Dictionary

Table 5: Complete algebra–physics correspondence.

Algebraic element	Realisation	Physical interpretation
Mode V	Vector in $\mathrm{Im}(\mathbb{H})$	Gravitational/visible field
Mode m	Vector in $\mathrm{Im}(\mathbb{H})$	Mass, inertia, confinement
Mode a	Vector in $\mathrm{Im}(\mathbb{H})$	Interaction, adjustment
Mode U	Real unit $1 \in \mathbb{H}$	Arena, background, witness
$\sigma = -1$	Left $\mathbb{H}$-action	Left chiral sector
$\sigma = +1$	Right $\mathbb{H}$-action	Right chiral sector
$\|m\|/\|V\|$	Norm ratio	Determines $\eta(n)/\alpha$
$I(n)$	Modal norm ratio	Scale G/α
$\eta(n) = \alpha I(n)$	Dimensionless product	Gravitational instability
$2^n = 1 + (2^n - 1)$	Identity decomposition	Witness vs. registered structure

3.1 Calibration

The algebraic norms become physical frequencies through:

$$\omega_N = \frac{m_p c^2}{\hbar}, \qquad R_V = \frac{c}{\omega_V} = 2^{n/2}\bar{\lambda}_p, \qquad \bar{\lambda}_p = \frac{\hbar}{m_p c}.$$

At $n = 4$:

$$\omega_V = \frac{\omega_p}{4}, \quad \omega_m = \frac{\sqrt{15}}{4}\omega_p, \quad R_V = 4\bar{\lambda}_p \approx 0.8412 \text{ fm}.$$

[HYPOTHESIS] The proximity of R_V to the CODATA 2018 proton charge radius (0.8414 ± 0.0019 fm) and the muonic-hydrogen value (0.84087 ± 0.00039 fm) is a falsifiable consistency check.

3.2 Gravitational instability parameter

The gravitational instability parameter is expressed here entirely in terms of the modal invariant $I(n)$ and the coupling α, without direct reference to G. (α is subsequently fixed by one calibration; see Section 4.3.)

Starting from the expression $G = \alpha R_V^2 a_m/m_p$, the gravitational interaction energy between two nucleons at separation $r = R_V$ is

$$U(R_V) = -G\frac{m_p^2}{R_V} = -\alpha R_V a_m m_p. \tag{1}$$

Substituting $R_V = 2^{n/2}\bar{\lambda}_p$ and $a_m = c^2/R_m = c^2\sqrt{1-2^{-n}}/\bar{\lambda}_p$:

$$U(R_V) = -\alpha\, 2^{n/2}\bar{\lambda}_p \cdot \frac{c^2\sqrt{1-2^{-n}}}{\bar{\lambda}_p} \cdot m_p = -\alpha\, c^2\, m_p\, 2^{n/2} \cdot \frac{\sqrt{2^n-1}}{2^{n/2}} = -\alpha\, c^2 m_p \sqrt{2^n-1}.$$

Using $c^2 m_p = \hbar\omega_N$:

$$U(R_V) = -\alpha\, \hbar\omega_N \sqrt{2^n-1}. \tag{2}$$

The field energy of the visible mode is $E_{\text{field}} = \hbar\omega_V = \hbar\omega_N 2^{-n/2}$. Hence

$$\boxed{\eta(n) := \frac{|U(R_V)|}{E_{\text{field}}} = \frac{\alpha\, \hbar\omega_N\sqrt{2^n-1}}{\hbar\omega_N\, 2^{-n/2}} = \alpha\, 2^{n/2}\sqrt{2^n-1} = \alpha\, I(n).} \tag{3}$$

[DERIVED] The result $\eta(n) = \alpha I(n)$ holds for all n. No reference to G appears; only α (a free parameter at this stage) and the modal invariant $I(n)$ enter. (Note: the identity $c^2 m_p = \hbar\omega_N$ was used to eliminate fundamental constants, leaving only α and $I(n)$.)

[DERIVED] Evaluating at $n = 4$:

$$\eta(4) = \alpha I(4) = \alpha\sqrt{240}.$$

Since the standard gravitational coupling satisfies $\alpha_G = Gm_p^2/(\hbar c) = \alpha I(4)$ (from the expression for G), we identify $\alpha_G = \eta(4)$.

[DERIVED] The Planck scale is the closure depth where the system becomes gravitationally self-referential, defined by $\eta(n_c) = 1$:

$$\alpha\, I(n_c) = 1 \quad \Longrightarrow \quad \alpha = \frac{1}{I(n_c)},$$

and therefore

$$\boxed{\alpha_G = \frac{I(4)}{I(n_c)} = \frac{\sqrt{240}}{I(n_c)}.}$$

The entire gravitational hierarchy reduces to determining n_c from first principles.

4 Non-relativistic Physical Framework

4.1 Relational distance and the inverse-square law

Postulate 4 (Relational difference elimination)**.** *Physical systems tend to reduce their mutual relational phase offset* $\Phi(r) = r/R_{\rm id}$*, where* $R_{\rm id} \equiv c/\omega_V$ *is the identity radius.*

Postulate 5 (Isotropic propagation)**.** *The harmonic closure condition selects no preferred spatial direction; the relational phase offset propagates isotropically.* ***[OPEN]*** *A derivation of this isotropy from Theorems 1–2 is an open problem.*

[DERIVED] Conservation of relational information over concentric surfaces:

$$\Delta\Phi(r) \cdot 4\pi r^2 = \text{const} \implies \Delta\Phi(r) = \Delta\Phi(R_{\rm id}) \cdot \frac{R_{\rm id}^2}{r^2}.$$

This is the *inverse-square law*, derived from conservation and isotropy, not assumed.

4.2 Gravitational potential

[HYPOTHESIS] The gravitational potential is the energy cost of maintaining a relational phase offset:

$$\phi(r) = -\alpha\,\omega_m^2 \int_r^\infty \Delta\Phi(r')\,dr' \cdot \frac{M}{m_p} = -\frac{\alpha\,\omega_m^2\,R_{\rm id}^2}{m_p} \cdot \frac{M}{r}.$$

[DERIVED] Identifying with Newton's potential $\phi_N = -GM/r$:

$$\boxed{G = \alpha\,\kappa\,\omega_m^2 = \alpha \cdot \frac{R_V^2}{m_p} \cdot a_m,}$$

where $\kappa = R_V^2 R_m/m_p$ is the specific relational closure volume and $a_m = c^2/R_m$ is the confinement centripetal acceleration. This separates G into three structurally independent factors:
1. α — relational transmission probability;
2. R_V^2/m_p — specific identity cross-section;
3. a_m — confinement acceleration (intrinsic coherence rate).

4.3 Calibration

[CALIBRATION] Setting $G = 6.67430 \times 10^{-11}$ m^{3}kg^{-1}s^{-2} (CODATA 2018) fixes $\alpha \approx 3.81 \times 10^{-40}$. This is the *single empirical input* to the non-relativistic framework.

Numerical verification:

$$\alpha = 3.8124 \times 10^{-40}, \quad R_V^2/m_p = 4.2309 \times 10^{-4}\ \text{m}^2/\text{kg}, \quad a_m = 4.1378 \times 10^{32}\ \text{m/s}^2.$$

Product $= 6.6743 \times 10^{-11}$ m^{3}kg^{-1}s$^{-2} = G$ (error $< 10^{-6}$).

4.4 Natural gravitational scale

[DERIVED] The product $\kappa\omega_m^2$ is constructible entirely from $m_p, c, \hbar$ and the $n = 4$ closure, without gravitational input. Using $\bar{\lambda}_p^3\omega_p^2 = \hbar c/m_p$ (exact):

$$\frac{G}{\alpha} = \kappa\omega_m^2 = I(4)\,\frac{\hbar c}{m_p^2} = \sqrt{240}\,\frac{\hbar c}{m_p^2} \approx 1.751 \times 10^{29}\ \text{m}^3\text{kg}^{-1}\text{s}^{-2}.$$

[DERIVED] The dimensionless coupling satisfies:

$$\alpha = \alpha_G \cdot \frac{\omega_V^2}{\omega_N\omega_m} = \frac{\alpha_G}{I(4)} = \frac{\alpha_G}{\sqrt{240}}.$$

4.5 Three centripetal accelerations of $n = 4$

$$a_N = \frac{c^2}{\overline{\lambda}_p} \approx 4.27 \times 10^{32}\ \mathrm{m/s^2}, \quad a_V = \frac{c^2}{R_V} \approx 1.07 \times 10^{32}\ \mathrm{m/s^2}, \quad a_m = \frac{c^2}{R_m} \approx 4.14 \times 10^{32}\ \mathrm{m/s^2}.$$

[DERIVED] Their ratios reproduce the modal frequency ratios: $a_V/a_N = 1/4$, $a_V/a_m = 1/\sqrt{15}$, and $\alpha_{\mathrm{geom}} = (1/4)(1/\sqrt{15}) = 1/\sqrt{240} = 1/I(4)$ is determined entirely by the $n = 4$ closure structure.

4.6 Geometric relation

[DERIVED] From $R_V = c/\omega_V$, $R_m = c/\omega_m$, $R_N = c/\omega_N$:

$$R_V^2 = I(n)\, R_N R_m.$$

This eliminates α from the modal geometry entirely.

5 Covariant Extension

*This part is preliminary. Results labelled **[DERIVED]** follow from postulates stated here; **[POSTULATE]** labels require further justification.*

5.1 Scalar field and covariant realisation

[POSTULATE] Define $\psi = m_p \phi_{\text{rel}}$ (mass dimensions, $\hbar = c = 1$). In static vacuum $\nabla^2 \psi = 0$.

[DERIVED] The operator $\times$ is realised covariantly: $(A \times R)^\mu = \epsilon^\mu{}_{\nu\rho\sigma} A^\nu R^\rho n^\sigma$, where n^μ is a unit timelike vector with $n^\mu \nabla_\mu \psi = 0$. This satisfies (P1)–(P3) automatically.

[POSTULATE] Projection 2-form: $P_{\mu\nu} = n_{[\mu} \nabla_{\nu]} \psi$. The double-projection identity is $(P_{\mu\nu} X^\nu) \nabla^\rho \psi \, P_{\rho\sigma} = \sigma (\nabla^\alpha \psi \nabla_\alpha \psi) X_\mu$.

[DERIVED] $P_{\mu\nu} P^{\mu\nu} = -\frac{1}{2} \nabla^\mu \psi \nabla_\mu \psi$ (see Appendix A).

5.2 Action and field equations

[POSTULATE] Total action:

$$S = \int d^4x \sqrt{-g} \left[\frac{\mathcal{R}}{16\pi G_0} + \frac{1-\beta}{2} \nabla^\mu \psi \nabla_\mu \psi - V(\psi) \right] + S_m,$$

where $\beta = 0$ in the weak-field limit; $\beta \neq 0$ is a speculative singularity regularisation.

[DERIVED] Einstein equations $G_{\mu\nu} = 8\pi G_0 (T^{(m)}_{\mu\nu} + T^{(\psi)}_{\mu\nu})$ follow by variation, with

$$T^{(\psi)}_{\mu\nu} = (1-\beta) \big(\nabla_\mu \psi \nabla_\nu \psi - \tfrac{1}{2} g_{\mu\nu} \nabla_\alpha \psi \nabla^\alpha \psi \big) - g_{\mu\nu} V(\psi).$$

5.3 Newtonian limit

[POSTULATE] Matter coupling: $\delta S_m / \delta \psi = \lambda \rho_{\text{rel}}$, $\rho_{\text{rel}} = \kappa \omega_m^2 \rho_m$.

In the non-relativistic limit $T^\mu{}_\mu \approx -\rho_m$, so this is equivalent to a scalar coupling to the trace of the energy-momentum tensor with a geometrically determined coefficient.

[DERIVED] In the weak-field static limit ($\beta = 0$): $\nabla^2 \psi = -\lambda \kappa \omega_m^2 \rho_m$. Identifying $\psi = -G m_p M / r$ gives $\lambda = -4\pi \alpha m_p$, reproducing the correct G.

[OPEN] Deriving the matter coupling from the algebraic structure (not postulating it) is the central open problem of this part.

5.4 Cosmological consequences

[HYPOTHESIS] Under a constant deceleration parameter $q = -\ddot{a}a/\dot{a}^2$ (phenomenological, low-redshift):

$$H(z) = H_0 (1+z)^{1+q}, \qquad d_L(z;q) = \frac{1+z}{H_0} \cdot \frac{1-(1+z)^{-q}}{q} \quad (q \neq 0).$$

For $q = -0.55$, deviations from ΛCDM reach $\sim 1\%$ at $z \sim 1$, within reach of LSST, Euclid, and DESI.

6 The Open Problem of α_G

6.1 Reduction to a single parameter

[DERIVED] With $n = 4$ and α calibrated: $\alpha_G = \alpha I(4) = \alpha\sqrt{240} \approx 5.906 \times 10^{-39}$. From Eq. (3), the Planck condition $\eta(n_c) = 1$ gives:

$$\boxed{\alpha_G = \frac{I(4)}{I(n_c)}.}$$

6.2 Three routes — all fail

[OPEN] All three natural routes fail; each failure is informative.

1. **Single-nucleon ratios.** Maximum suppression $\sim (\omega_V/\omega_N)^2(\omega_V/\omega_m)^2 \sim 10^{-2}$. Gap to α_G: 37 orders of magnitude.
2. **Collective closure.** $\eta(N) = \alpha_G N^{5/3} = 1$ gives $N_{\text{crit}} \approx 8.7 \times 10^{22}$ with no privileged physical interpretation.
3. **Covariant self-consistency.** The condition $\eta = 1$ at the Planck scale yields α_G on both sides — circular.

Conclusion. The origin of α_G lies outside the four-mode structure of a single nucleon. Determining n_c from first principles is the central open problem.

6.3 The E_8 program and its status

Step	Construction	Mathematical content	Status
A	$\{V, m, a\}$, $\sigma = -1$	$\mathfrak{so}(3) \cong A_1$	**Verified**
B	$\sigma = -1 \oplus \sigma = +1$	$\mathfrak{so}(4) \cong D_2$	**Verified**
C	Cayley–Dickson $\mathbb{H} \oplus \mathbb{H}$	$\mathbb{O}$	Pending
D	$\mathrm{Aut}(\mathbb{O})$ as vector space	$\mathrm{SO}(8) \cong D_4$	Pending
E	$D_4 \times D_4 \hookrightarrow E_8$	24+24+192=240 roots	Pending

Steps C–E follow established mathematics but require translation into modal language. Steps A–B are verified by explicit 4×4 matrix computation (Theorems 1–2).

[OPEN] Completing Steps C–E would replace the arithmetic observation $I(4)^2 = 240$ with a structural theorem deriving 240 as the root count of E_8 from the modal algebra.

Why the E_8 program does not close α_G. Even a complete chain A–E would establish why $I(4) = \sqrt{240}$, but not the value of n_c. The Coxeter number of E_8 is $h = 30$, the rank is 8, the dimension is 248 — none of these equals 2^{127}. The E_8 program and the α_G problem are distinct open questions.

6.4 A numerical hint

[SPECULATIVE] A purely arithmetic observation (not a derivation):

$$\alpha_G \approx 2^{-127}\left(1 + \frac{1}{240}\right), \quad \text{error } 0.08\%.$$

In the language of $\alpha_G = I(4)/I(n_c)$ with $I(n) \approx 2^n$: $\alpha_G \approx 2^{4-n_c}$, giving $n_c \approx 131 = 127 + 4$. The number 127 is the Mersenne prime $2^7 - 1$. The correction $1/240 = 1/I(4)^2$ reflects the deviation of $I(4)$ from 2^4.

Why this is not a derivation: (a) the error 0.08% exceeds the experimental uncertainty of α_G ($\approx 0.002\%$), so the formula is not exact; (b) $n_c = 131$ has no derivation within the generative algebra; (c) $131 = 127 + 4$ is a numerical observation, not a theorem. Recorded in the spirit of Dirac's large-number hypothesis[1], as a possible pointer to deeper structure.

[1] Unlike Dirac's approach, here the large number 2^{127} emerges from the closure hierarchy ($n_c \approx 131$) rather than from cosmological coincidence.

[OPEN] Determine n_c from the generative algebra without gravitational input. Possible routes: a global consistency condition $\prod_n I(n) = \text{const}$; connection to the E_8 Weyl group or Coxeter number; collective-closure argument beyond single-nucleon structure.

7 Testable Predictions

7.1 Weak equivalence principle violations

[HYPOTHESIS] If ω_m or $\kappa = R_V^2 R_m/m_p$ varies with nuclear composition — e.g., via binding energy per nucleon B — the framework predicts WEP violations. The Eötvös parameter is:

$$\eta_{A,B} \approx \frac{\Delta\omega_m^2}{\omega_m^2} + 2\frac{\Delta R_V}{R_V} + \frac{\Delta R_m}{R_m}.$$

If ω_m^2 scales linearly with B: $\eta \approx \beta\,\Delta B/B + \ldots$

Falsification condition. WEP violations should correlate with binding energy per nucleon, not only atomic mass. Current MICROSCOPE limit: $\eta \lesssim 10^{-15}$. Falsified if no such correlation appears at STE-QUEST precision.

Next concrete step. Map nuclear binding energy to $\Delta\omega_m$ and ΔR_m using nuclear structure data to produce a quantitative prediction.

7.2 Cosmological deviations from ΛCDM

[DERIVED] Under the constant-q approximation, $d_L(z;q)$ deviates from ΛCDM by 1–3% at $z \sim 1$–2, testable with Type Ia supernovae and gravitational-wave standard sirens (LSST, Euclid, DESI).

7.3 Scale-dependent gravity

[SPECULATIVE] If closure is partial in diffuse systems, $\kappa(r) = R_V^2(r)R_m(r)/m_p$ becomes scale-dependent, giving $G_{\text{eff}}(r) = \alpha\kappa(r)\omega_m^2$. Rotation-curve anomalies should then correlate with informational density $\omega_m^2\rho_m$. *Caveat: the r-dependence of κ is not derived from the postulates; this is a program note.*

8 Discussion

8.1 Comparison with emergent-gravity frameworks

Table 6: Comparison with emergent-gravity approaches.

Approach	Source of gravity	G status	$1/r^2$ law	Algebra
Newton/Einstein	Mass/energy	Fundamental	Assumed	—
Verlinde	Entropy gradient	Proportionality	Assumed	—
Jacobson	Horizon thermodynamics	Via $S \propto A/G$	Assumed	—
Padmanabhan	Horizon thermodynamics	Emergent (qual.)	Assumed	—
This work	Phase-offset density	$G = \alpha R_V^2 a_m/m_p$	Derived	$\mathfrak{so}(4)$

The critical distinction is that all prior approaches assume the $1/r^2$ form from the outset; the present framework derives it from conservation and isotropy. Moreover, G is expressed in terms of hadronic-scale physics (R_V, R_m, m_p) rather than absorbed into entropy definitions.

8.2 The mechanism

Why does the internal structure of a proton affect a trajectory 15 light-minutes away? Within this framework, *distance is a relational quantity.* Each nucleon projects outward via its visible cross-section R_V^2, establishing a phase-offset field. The inverse-square decay follows from isotropy and information conservation — the same logic that governs any conserved quantity on a sphere.

8.3 Other manifestations

[SPECULATIVE] The four-mode structure should describe any system where a relational phase offset develops and is eliminated. Qualitative examples: atomic transitions (visible mode → photon), lightning (electric phase difference → discharge), aurora. These require quantitative development.

8.4 Limitations and future work

Current limitations: weak-field regime only; α calibrated (one free parameter); isotropy not derived from algebra; E_8 chain Steps C–E pending; matter coupling in Part II postulated; n_c derivation open.

Priority future work:

1. Complete Steps C–E of the E_8 program.
2. Derive isotropy from Theorems 1–2.
3. Calculate the WEP prediction from nuclear binding data.
4. Determine n_c from the algebraic hierarchy.
5. Develop the strong-field tensor formulation.

9 Conclusion

We have derived Newtonian gravity — and its covariant extension — from relational projection principles in which neither the inverse-square law nor Newton's constant are assumed.

Established algebraic results:
- Modes $\{V, m, a\}$ with $\sigma = -1$ generate $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ (Theorem 1).
- Both orientations generate $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ (Theorem 2); σ is the chiral distinction.
- $I(4)^2 = 240 = \#\text{roots}(E_8)$ (arithmetic, exact).
- $\eta(n) = \alpha I(n)$ derived from modal invariants (Eq. 3).
- $\alpha_G = I(4)/I(n_c)$; open: determine n_c.

Core physical results:
- Inverse-square law derived from conservation and isotropy.
- $G = \alpha(R_V^2/m_p)a_m$: three structurally independent factors.
- Natural scale $G/\alpha = \sqrt{240}\,\hbar c/m_p^2$, no gravitational input required.

Open problems, in priority order:
1. Derive α_G (determine n_c without gravitational input).
2. Complete the E_8 chain (Steps C–E).
3. Derive isotropy from the quaternionic structure.
4. Quantify the WEP prediction from nuclear data.

Gravity, like the size of the proton, the stability of atoms, the expansion of the cosmos, and the lightning in the sky, may be a manifestation of relational geometry seeking harmonic closure. Whether this program closes depends on whether n_c can be derived without gravitational input. That is the question.

A Contraction of $P_{\mu\nu}$

From $P_{\mu\nu} = n_{[\mu}\nabla_{\nu]}\psi$ and $n^\mu\nabla_\mu\psi = 0$:

$$\begin{aligned} P_{\mu\nu}P^{\mu\nu} &= \tfrac{1}{4}(n_\mu\nabla_\nu\psi - n_\nu\nabla_\mu\psi)(n^\mu\nabla^\nu\psi - n^\nu\nabla^\mu\psi) \\ &= \tfrac{1}{4}(-2\nabla_\mu\psi\nabla^\mu\psi) = -\tfrac{1}{2}\nabla^\mu\psi\nabla_\mu\psi. \qquad \square \end{aligned}$$

B Geometric relation $R_V^2 = I(n)R_N R_m$

With $R_V = 2^{n/2}\bar{\lambda}_p$, $R_m = \bar{\lambda}_p/\sqrt{1-2^{-n}}$, $R_N = \bar{\lambda}_p$:

$$I(n)R_N R_m = 2^{n/2}\sqrt{2^n-1}\cdot\bar{\lambda}_p\cdot\frac{\bar{\lambda}_p}{\sqrt{(2^n-1)/2^n}} = 2^n\bar{\lambda}_p^2 = R_V^2. \qquad \square$$

C Generator matrices and commutator verification

The six 4×4 matrices (acting on the quaternion basis $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$) are listed in Theorems 1 and 2. Direct computation confirms: $[L_V, L_m] = 2L_a$, $[L_m, L_a] = 2L_V$, $[L_a, L_V] = 2L_m$ ($\mathrm{so}(3)_L$ sector); $[R_i, R_j] = 2R_k$ ($\mathrm{so}(3)_R$ sector); $[L_i, R_j] = 0$ for all six pairs (decoupling). All verifications were carried out by explicit symbolic computation.

D Glossary of symbols

Symbol	Definition
$\hat{\aleph}$	Normalised reference state
R	Differentiation marker, $\|R\| \in [0,1]$
$\sigma \in \{+1,-1\}$	Binary orientation (left/right $\mathbb{H}$-action)
$\{U, V, m, a\}$	Four generated modes
n	Binary closure depth ($n = 4$ for nucleon)
$I(n) = 2^{n/2}\sqrt{2^n-1}$	Modal invariant
R_V, R_m, R_N	Identity, confinement, Compton radii
$\kappa = R_V^2 R_m/m_p$	Specific relational closure volume
$\alpha \approx 3.81\times 10^{-40}$	Relational coupling (one calibration)
$\alpha_G = Gm_p^2/(\hbar c)$	Standard gravitational coupling
$a_m = c^2/R_m$	Confinement centripetal acceleration
$\eta(n) = \alpha I(n)$	Gravitational instability parameter
n_c	Planck closure depth ($\eta(n_c) = 1$); open

Computational disclosure

Version 9.1 was developed with iterative algebraic verification assisted by large language models (Anthropic Claude). These were used for matrix commutator checks, dimensional consistency verification, and exploration of the E_8 connection. All mathematical claims were independently verified by the author. The verification of Theorems 1–2 was carried out by explicit 4×4 matrix computation; source code is available from the author on request.

References

[1] E. Tiesinga, P. J. Mohr, D. B. Newell, B. N. Taylor, *CODATA recommended values of the fundamental physical constants: 2018*, Rev. Mod. Phys. **93**, 025010 (2021).

[2] E. Verlinde, *On the Origin of Gravity and the Laws of Newton*, JHEP **04**, 029 (2011).

[3] T. Jacobson, *Thermodynamics of Spacetime: The Einstein Equation of State*, Phys. Rev. Lett. **75**, 1260 (1995).

[4] T. Padmanabhan, *Thermodynamical Aspects of Gravity: New Insights*, Rep. Prog. Phys. **73**, 046901 (2010).

[5] P. A. M. Dirac, *A New Basis for Cosmology*, Proc. R. Soc. A **165**, 199 (1938).

[6] P. Touboul et al. (MICROSCOPE Collaboration), *MICROSCOPE Mission: First Results of a Space Test of the Equivalence Principle*, Phys. Rev. Lett. **119**, 231101 (2017).

[7] M. Milgrom, *A Modification of the Newtonian Dynamics as a Possible Alternative to the Hidden Mass Hypothesis*, Astrophys. J. **270**, 365 (1983).

[8] A. Hurwitz, *Über die Komposition der quadratischen Formen von beliebig vielen Variablen*, Nachr. Ges. Wiss. Göttingen, 309 (1898).

[9] J. H. Conway, D. A. Smith, *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*, A K Peters (2003).

[10] J. F. Adams, *Lectures on Exceptional Lie Groups*, University of Chicago Press (1996).

[11] R. Pohl et al., *The size of the proton*, Nature **466**, 213 (2010).

[12] B. Famaey, S. McGaugh, *Modified Newtonian Dynamics (MOND): Observational Phenomenology and Relativistic Extensions*, Living Rev. Relativ. **15**, 10 (2012).

[13] L. D. Mata Sánchez, *Relational Geometry: prior versions v4.0–v9.0*, unpublished preprints (2025–2026).